

KRISHAN KUMAR

Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Developer with more than 4 years of experience designing, developing, and deploying scalable web applications. Proficient in MERN, Java Springboot, Next.js, Vue.js, and relational & NoSQL databases. Skilled in cloud deployment (Docker, Kubernetes, AWS S3), CI/CD, and Agile methodologies. Proven track record of improving system performance by up to 60%, boosting user engagement by 40% and delivering enterprise-grade solutions. Passionate about building efficient, maintainable, and user-centric software.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Java
Frontend: React.js, Next.js, Vue, Redux, Vuex, HTML5, CSS3, Vite, Flutter, Payload CMS
Design Frameworks: Material-UI, Bootstrap, ApexCharts, Vuetify, DevExtreme
Backend: Node.js, Express.js, REST services, Microservices, Nestjs, Springboot
Databases: MongoDB, PostgreSQL, MySQL, SQL
Tools & DevOps: Docker, Kubernetes, Git, GitHub Actions, Websockets, Redis, Socket.IO
Testing & Debugging: Jest, Chrome DevTools, Unit Testing/Integration Testing
Amazon Services: EC2, S3, ElastiCache, Route 53, Secrets Manager
Methodologies: Agile (Kanban, Zoho Sprints, JIRA), SDLC, OOP, data structures and algorithms

EXPERIENCE AND PROJECT BASED CONTRIBUTIONS

ULTRASHIELD TECHNOLOGIES

Noida, UP

MERN Stack Developer

11/2025 - Present

- Contributed to the improvements in backend like advanced filters, redis queue management using Amazon ElastiCache, app deployment on AWS EC2 using Github action workflow.

SHINE WEB SERVICES

Mohali, PB

MERN Stack Developer

05/2022 - 10/2025

- Contributed to the development of **Healthcore (Patient Management System)** focusing on appointment scheduling, patient management and engagement, secure doctor-patient interactions and improving healthcare workflows.

PROJECTS

[HealthCore \(Patient Management\)](#)

Tech Stack : Vue.js, Bootstrap, Vuetify, Pinia, DevExtreme

- Implemented Appointment Scheduler with DevExtreme, improving doctor-patient scheduling efficiency.
- Built Doctor Management Module for admins to manage patient records, documents, and appointments.
- Developed Patient Portal for doctors to track profiles, tasks, and medical history.
- Integrated SSO login for patients to submit complaints via simplified forms, enhancing user experience.

[Pleezr \(Dating Application\)](#)

Tech Stack : TypeScript, Node.js MongoDB Express.js, Socket.IO, Stripe, Twilio, React.js

- Query optimizations, advanced filters reduce response load times by 60% and deployment on EC2 using Github Actions enabled setup CICD.
- Gamification features with badges and challenges, referral system, profile boost functionality, Aadhar verification, content moderation, Firebase Admin for push notifications.
- Stripe payment gateway integration with subscription management, metered usage tracking, cron workers for daily reset and maintenance, webhook handling for payment processing.
- File uploads to S3 enabled smooth retrieval and improved file maintenance and setup custom scripts too to retrieve data from excel sheets.

TukTukLove (Dating Application)

Tech Stack : TypeScript, Node.js, MongoDB, Express.js, Socket.IO, Cashfree, Twilio, Agora, React.js

- Microservices-ready Node.js/Express backend with TypeScript, 20+ modular components including user matching, messaging, voice calls, subscriptions, MongoDB database with Redis caching and AWS S3 storage, Socket.io for real-time features, RESTful APIs with Swagger documentation
- Stripe, Razorpay, Cashfree payment gateway integration with subscription management, metered usage tracking, cron workers for daily reset and maintenance, webhook handling for payment processing.
- Multi-layer rate limiting, brute force protection, JWT authentication with OTP verification, Helmet security headers, input sanitization, IP filtering, CORS configuration, NoSQL injection and XSS prevention
- React TypeScript with Material-UI components, Redux Toolkit for state management, admin panel with CMS and analytics dashboards, user management, verification workflows, push notification campaigns, support ticket system
- PM2 clustering for production deployment, BullMQ job queue management, MongoDB change streams for cache synchronization, environment-based configuration management, Agora voice calling integration
- Gamification features with badges and challenges, referral system, profile boost functionality, Aadhar verification, content moderation, Firebase Admin for push notifications.

Curotiva (Wellness Application)

Tech Stack : Java Springboot, Websockets, Amazon Elasticache, Agora, Twilio, MongoDB, Cashfree

- Led backend optimization and modularization efforts, resulting in a 40% improvement in code maintainability and development velocity.
- Designed and deployed a WebSocket-based real-time communication architecture capable of scaling to 50,000+ users, separating chat and queue updates for improved throughput and reliability.
- Implemented a distributed, Redis-backed queue management system using Amazon ElastiCache, reducing matchmaking response time by 40% under high traffic.
- Built an end-to-end therapist booking and queue assignment flow with real-time session control and therapist-driven user selection.
- Implemented real-time, usage-based billing logic (per-minute), improving billing accuracy by 35%.
- Delivered production-ready Agora-based audio calling with low latency and high concurrency support.

Genic Assets (Asset Management Software)

Tech Stack : MERN

- Mongoose and MongoDB conjunction enhanced database interactivity and usage of Aggregation pipelines, facilitating data processing and improving performance by 50% during querying.
- Upgradation in company AMS results in 40% smoother handling of inventory and increased customer satisfaction.

EDUCATION

Bachelor of Computer Application (**BCA**)

Kurukshetra University, HR | 2016-2019

CERTIFICATIONS

Frontend Developer (React)

Javascript (Basic) Certificate

SQL (Basic) Certificate

HackerRank (05762F05A7DD)

HackerRank (FBBC778792AF)

HackerRank (CDB6D99F9273)